

PROJECT REPORT

Data Cleaning & Preparation

DecodeLabs Data Analytics Internship
Industrial Training Kit | Project 1 | Batch 2026

Project Title	Data Cleaning & Preparation
Organization	DecodeLabs
Program	Data Analytics Internship – Industrial Training Kit
Batch	2026
Tool Used	Python + Pandas
Date	31 August 2026

1. Introduction

This report documents the complete process and outcomes of Project 1 under the DecodeLabs Data Analytics Internship. The primary objective was to clean and prepare a raw e-commerce order dataset so that it becomes a reliable foundation for future analysis, modeling, and dashboarding work.

According to industry estimates, poor data quality costs organizations an average of \$12.9 million annually. Data cleaning is therefore not a peripheral task but the foundation of trustworthy business intelligence. This project emphasizes Data Integrity over visual analytics.

2. Project Objectives

- Identify and handle missing or null values using strategic imputation.
- Detect and eliminate duplicate records while preserving unique identifiers.
- Standardize data formats (dates, numeric precision, text consistency).
- Achieve 0% error rate on unique identifiers and date formats (Verification Gate).
- Deliver a fully reproducible cleaning pipeline.

3. Dataset Description

The source file Dataset for Data Analytics.xlsx contains 1,200 order records spanning from 1 January 2023 to 30 June 2025. Each record represents a single customer order with the following attributes:

Column	Description
OrderID	Unique order identifier (primary key)
Date	Order date
CustomerID	Customer identifier
Product	Product category (Phone, Laptop, Monitor, etc.)
Quantity	Units ordered (1–5)
UnitPrice	Price per unit
ShippingAddress	Delivery address
PaymentMethod	Payment mode used
OrderStatus	Current status of the order
TrackingNumber	Shipment tracking ID
ItemsInCart	Total items in the cart
CouponCode	Discount code applied (if any)
ReferralSource	Marketing channel of acquisition
TotalPrice	Final order value

4. Methodology

The cleaning process followed three structured phases aligned with professional data engineering standards:

4.1 Phase 1 – Strategic Imputation

Missing values were identified exclusively in the CouponCode column (309 records, 25.75%). Instead of deleting these rows (listwise deletion), the missing values were imputed with the string “No Coupon”. This approach preserves the full sample size and statistical power, as recommended in the project guidelines.

4.2 Phase 2 – Integrity Audit

A complete uniqueness and consistency audit was performed. No duplicate rows, duplicate OrderIDs, or duplicate TrackingNumbers were found. Additionally, the calculated field Quantity × UnitPrice matched the provided TotalPrice for every record (zero mismatches).

4.3 Phase 3 – Standardization

All text columns were trimmed of potential whitespace. Monetary columns (UnitPrice, TotalPrice) were rounded to two decimal places for reporting consistency. The Date column was validated as proper datetime objects ready for ISO 8601 formatting.

5. Results & Verification Gate

After applying the cleaning pipeline, the dataset successfully met all Verification Gate criteria required to unlock Project 2:

Metric	Before	After	Status
Missing CouponCode	309	0	PASS
Duplicate OrderIDs	0	0	0% Error
Invalid Dates	0	0	0% Error
TotalPrice mismatches	0	0	PASS
Records preserved	1,200	1,200	100%

6. Deliverables

- Dataset for Data Analytics.xlsx – Original raw dataset
- cleaned_dataset.xlsx – Final analysis-ready dataset
- data_cleaning.py – Fully reproducible Python cleaning script
- Project_Report.docx – This formal project report
- README.md – Professional GitHub documentation

7. Conclusion

Project 1 successfully transformed a raw order dataset into a clean, documented, and reproducible data asset. All original records were retained, unique identifiers remain free of duplicates, and date formats are fully standardized. The Verification Gate criteria have been met with a 0% error rate.

This work establishes a solid foundation for subsequent projects involving exploratory analysis, visualization, and predictive modeling. The emphasis on documentation and reproducibility reflects professional standards expected in real-world data teams.